

Pontryagin-Based Solver with Universally Smoothed Hamiltonian, Adaptive ΔT , and PA-Bundle Refinement

March 9, 2026

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1D Example: Hypersensitive Optimal Control

The Followign example is taken from Karlsson et. al. [1]:

Hypersensitive optimal control

Minimize

$$\int_0^{25} \left(X(t)^2 + \alpha(t)^2 \right) dt + \gamma (X(25) - 1)^2 \quad (1)$$

subject to

$$X'(t) = -X(t)^3 + \alpha(t), \quad 0 < t \leq 25, \quad (2)$$

$$X(0) = 1. \quad (3)$$

for $\gamma = 1e6$. The Hamiltonian is then given by

$$H(x, \lambda) = \min_{\alpha} \{ -\lambda x^3 + \lambda \alpha + x^2 + \alpha^2 \} = -\lambda x^3 - \frac{\lambda^2}{4} + x^2 \quad (4)$$

$$H_{\lambda}(x, \lambda) = -x^3 - \frac{\lambda}{2} \quad (5)$$

$$H_x(x, \lambda) = -3\lambda x^2 + 2x \quad (6)$$

Numerical Results: Explicit Hamiltonian

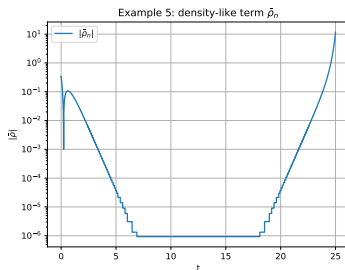
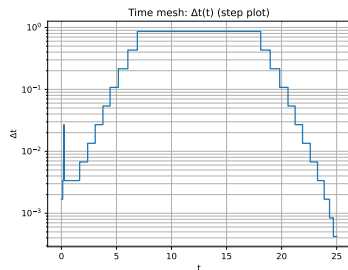
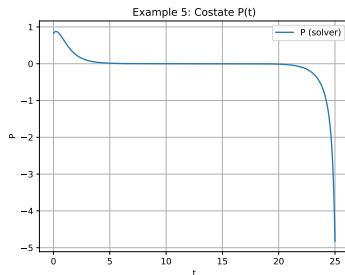
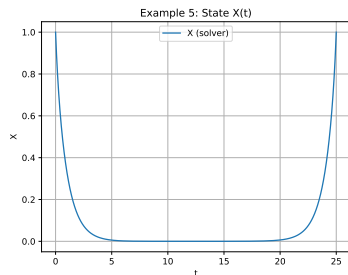


Figure: Example 5 results for $\varepsilon_{\text{time}} = 1e-2$.

Numerical Results: Explicit Hamiltonian

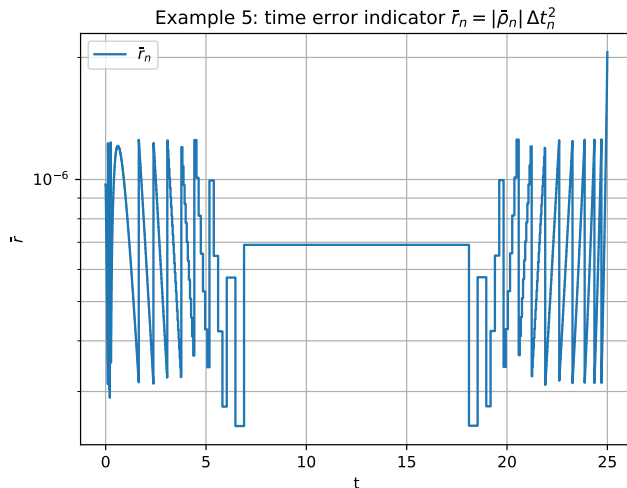


Figure: Error indicators $\bar{r}_n$ of the discrete optimization problem (7) for $\varepsilon_{\text{time}} = 1e - 2$.

Numerical Results: Explicit Hamiltonian

#	N	M	$dt_{\min}$	$dt_{\max}$	δ	N.lt	res	η_{time}	η_{PA}	η_{δ}	action
00	29	3	8.62e-01	8.62e-01	2e-02	6	1.88e-10	5.90e-01/3.45e-04	2.33e+01/1e-02	2.17e-01/1e-02	refine_time (m=8)
01	37	3	4.31e-01	8.62e-01	2e-02	5	7.27e-11	3.20e-01/2.70e-04	2.28e+01/1e-02	1.99e-01/1e-02	refine_time (m=16)
02	53	3	2.16e-01	8.62e-01	2e-02	4	2.02e-10	1.98e-01/1.89e-04	2.25e+01/1e-02	1.96e-01/1e-02	refine_time (m=31)
03	84	3	1.08e-01	8.62e-01	2e-02	4	1.27e-10	8.33e-02/1.19e-04	2.24e+01/1e-02	1.94e-01/1e-02	refine_time (m=59)
04	143	3	5.39e-02	8.62e-01	2e-02	4	1.21e-10	2.68e-02/6.99e-05	2.24e+01/1e-02	1.93e-01/1e-02	refine_time (m=102)
05	245	3	2.69e-02	8.62e-01	2e-02	3	4.22e-09	7.55e-03/4.08e-05	2.24e+01/1e-02	1.92e-01/1e-02	refine_time (m=169)
06	414	3	1.35e-02	8.62e-01	2e-02	3	2.03e-10	2.00e-03/2.42e-05	2.24e+01/1e-02	1.92e-01/1e-02	refine_time (m=271)
07	685	3	6.73e-03	8.62e-01	2e-02	3	1.91e-10	5.14e-04/1.46e-05	2.24e+01/1e-02	1.92e-01/1e-02	refine_time (m=374)
08	1059	3	3.37e-03	8.62e-01	2e-02	3	2.12e-11	1.30e-04/9.44e-06	2.24e+01/1e-02	1.92e-01/1e-02	refine_time (m=438)
09	1497	3	1.68e-03	8.62e-01	2e-02	3	1.03e-10	3.28e-05/6.68e-06	2.24e+01/1e-02	1.92e-01/1e-02	refine_time (m=490)
10	1987	3	8.42e-04	8.62e-01	2e-02	2	3.38e-09	8.23e-06/5.03e-06	2.24e+01/1e-02	1.92e-01/1e-02	refine_time (m=534)
11	2521	3	4.21e-04	8.62e-01	2e-02	2	8.57e-10	2.06e-06/3.97e-06	2.24e+01/1e-02	1.92e-01/1e-02	STOP

Numerical Results: Numeric Hamiltonian

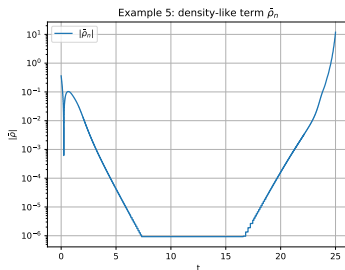
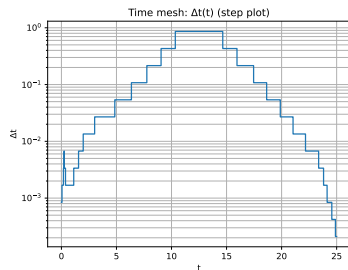
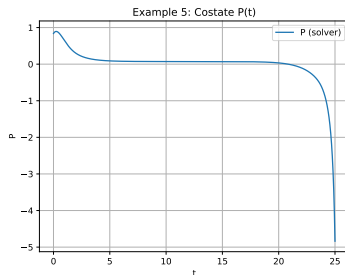
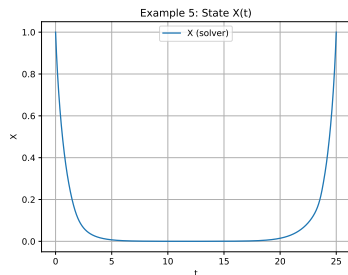


Figure: Example 5 results for $\epsilon_{\text{time}} = 1e-2$, $\epsilon_{\text{PA}} = 1e-2$, $\epsilon_{\delta} = 1e-2$, $\delta_0 = 0.02$.

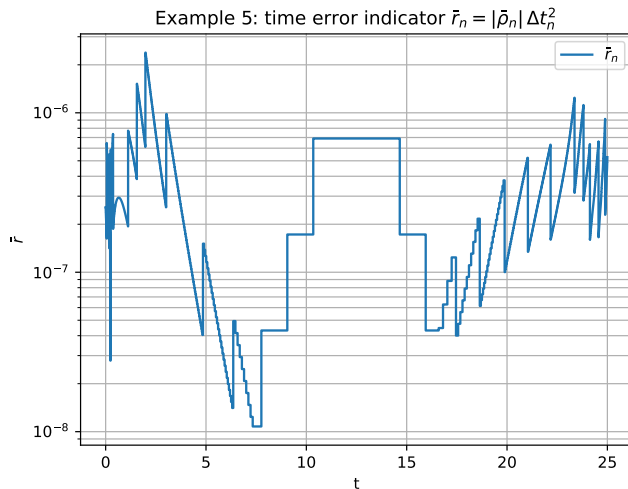


Figure: Error indicators $\bar{r}_n$ of the discrete optimization problem (7) for $\varepsilon_{\text{time}} = 1e - 2$, $\varepsilon_{\text{PA}} = 1e - 2$, $\varepsilon_{\delta} = 1e - 2$, $\delta_0 = 0.02$

Numeric Results: Numeric Hamiltonian

#	N	M	$dt_{\min}$	$dt_{\max}$	δ	N.lt	res	η_{time}	η_{PA}	η_{δ}	action
00	29	7	8.62e-01	8.62e-01	2e-02	9	1.18e-10	2.25e-01/3.45e-04	3.43e-01/1e-02	2.39e-01/1e-02	refine_time (m=13)
01	42	7	4.31e-01	8.62e-01	2e-02	13	1.32e-10	4.77e-01/2.38e-04	6.70e-01/1e-02	2.29e-01/1e-02	refine_time (m=26)
02	68	7	2.16e-01	8.62e-01	2e-02	8	7.10e-11	2.18e-01/1.47e-04	5.85e-01/1e-02	2.29e-01/1e-02	refine_time (m=43)
03	111	7	1.08e-01	8.62e-01	2e-02	5	7.82e-11	9.91e-02/9.01e-05	5.61e-01/1e-02	2.27e-01/1e-02	refine_time (m=71)
04	182	7	5.39e-02	8.62e-01	2e-02	9	4.57e-11	4.21e-02/5.49e-05	5.95e-01/1e-02	2.26e-01/1e-02	refine_time (m=100)
05	282	7	2.69e-02	8.62e-01	2e-02	5	9.07e-11	1.13e-02/3.55e-05	5.95e-01/1e-02	2.26e-01/1e-02	refine_time (m=157)
06	439	7	1.35e-02	8.62e-01	2e-02	7	5.05e-10	2.98e-03/2.28e-05	6.01e-01/1e-02	2.26e-01/1e-02	refine_time (m=209)
07	648	7	6.73e-03	8.62e-01	2e-02	6	5.82e-10	7.61e-04/1.54e-05	6.03e-01/1e-02	2.25e-01/1e-02	refine_time (m=306)
08	954	7	3.37e-03	8.62e-01	2e-02	4	8.91e-11	1.92e-04/1.05e-05	6.03e-01/1e-02	2.26e-01/1e-02	refine_time (m=392)
09	1346	7	1.68e-03	8.62e-01	2e-02	3	1.70e-09	4.81e-05/7.43e-06	6.03e-01/1e-02	2.26e-01/1e-02	refine_time (m=537)
10	1883	7	8.42e-04	8.62e-01	2e-02	3	1.58e-10	1.20e-05/5.31e-06	6.03e-01/1e-02	2.26e-01/1e-02	refine_time (m=539)
11	2422	7	4.21e-04	8.62e-01	2e-02	3	1.17e-10	3.01e-06/4.13e-06	6.03e-01/1e-02	2.26e-01/1e-02	add_plane
12	2422	8	4.21e-04	8.62e-01	2e-02	5	1.10e-10	2.18e-06/4.13e-06	5.05e-01/1e-02	2.26e-01/1e-02	add_plane
13	2422	9	4.21e-04	8.62e-01	2e-02	4	1.87e-10	2.49e-06/4.13e-06	4.87e-01/1e-02	2.26e-01/1e-02	add_plane
14	2422	10	4.21e-04	8.62e-01	2e-02	4	1.44e-10	2.44e-06/4.13e-06	4.83e-01/1e-02	2.27e-01/1e-02	add_plane
15	2422	11	4.21e-04	8.62e-01	2e-02	5	1.49e-12	3.36e-06/4.13e-06	3.90e-01/1e-02	2.31e-01/1e-02	add_plane
16	2422	12	4.21e-04	8.62e-01	2e-02	6	4.78e-09	1.41e-05/4.13e-06	1.17e-01/1e-02	2.51e-01/1e-02	refine_time (m=680)
17	3102	12	2.10e-04	8.62e-01	2e-02	3	2.17e-10	3.54e-06/3.22e-06	1.17e-01/1e-02	2.51e-01/1e-02	refine_time (m=420)
18	3522	12	2.10e-04	8.62e-01	2e-02	2	4.97e-09	8.88e-07/2.84e-06	1.17e-01/1e-02	2.51e-01/1e-02	add_plane
19	3522	13	2.10e-04	8.62e-01	2e-02	3	1.54e-10	9.11e-07/2.84e-06	1.15e-01/1e-02	2.52e-01/1e-02	add_plane
20	3522	14	2.10e-04	8.62e-01	2e-02	3	9.95e-11	9.11e-07/2.84e-06	1.12e-01/1e-02	2.52e-01/1e-02	add_plane
21	3522	15	2.10e-04	8.62e-01	2e-02	3	1.47e-10	9.09e-07/2.84e-06	1.11e-01/1e-02	2.53e-01/1e-02	add_plane
22	3522	16	2.10e-04	8.62e-01	2e-02	4	9.02e-11	1.28e-06/2.84e-06	9.31e-02/1e-02	2.58e-01/1e-02	add_plane
23	3522	17	2.10e-04	8.62e-01	2e-02	3	1.64e-10	1.26e-06/2.84e-06	8.88e-02/1e-02	2.60e-01/1e-02	add_plane
24	3522	18	2.10e-04	8.62e-01	2e-02	5	5.26e-11	2.37e-06/2.84e-06	2.28e-02/1e-02	3.65e-01/1e-02	add_plane

1D Example: Simple Optimal Control Problem

Simple Optimal Control Problem

Minimize

$$\int_0^1 X(t)^{10} dt \quad (7)$$

subject to

$$X'(t) = \alpha(t) \in [-1, 1], \quad 0 < t \leq 1, \quad (8)$$

$$X(0) = 0.5 \quad (9)$$

The Hamiltonian is then given by

$$H(x, \lambda) := \min_{\alpha \in [-1, 1]} \{\lambda \alpha + x^{10}\} = -|\lambda| + x^{10}. \quad (10)$$

The exact solution (without regularization) can be written as

$$X^*(t) = \begin{cases} 0.5 - t, & t \in [0, 0.5], \\ 0, & \text{otherwise,} \end{cases} \quad \alpha^*(t) = \begin{cases} -1, & t \in [0, 0.5], \\ 0, & \text{otherwise.} \end{cases} \quad (11)$$

$$u(0, x_0) = \int_0^1 X^*(t)^{10} dt = \frac{0.5^{11}}{11} \quad (12)$$

H_δ

For the planes

$$g_i(p, x, t) := p^\top f(x, u^{(i)}, t) + \ell(x, u^{(i)}, t). \quad (13)$$

The PA surrogate is

$$\bar{H}(p, x, t) := \min_{1 \leq i \leq m} g_i(p, x, t). \quad (14)$$

The function 'smoothing.py' replaces the hard minimum by a soft-min / log-sum-exp smooth approximation:

$$H_\delta(p, x, t) := -\delta \log \left(\sum_{i=1}^m \exp \left(-\frac{g_i(p, x, t)}{\delta} \right) \right), \quad \delta > 0. \quad (15)$$

H_δ used in Karlsson et. al. [1]

$$H_\delta(x, \lambda) = -\sqrt{\lambda^2 + \delta^2} + x^{10} \quad (16)$$

Numerical Results: Numeric Hamiltonian

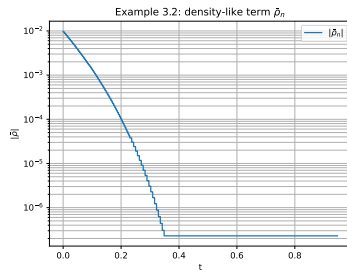
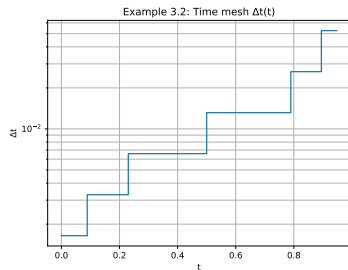
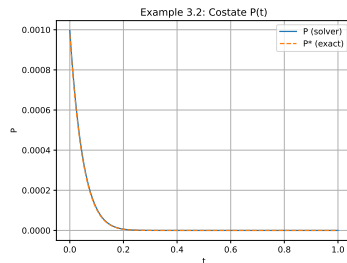
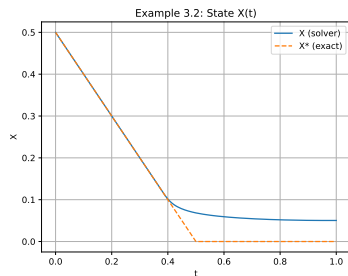


Figure: Example 5 results for $\varepsilon_{\text{time}} = 1e-5$, $\varepsilon_{\text{PA}} = 1e-4$, $\varepsilon_{\delta} = 1e-10$, $\delta_0 = 0.02$.

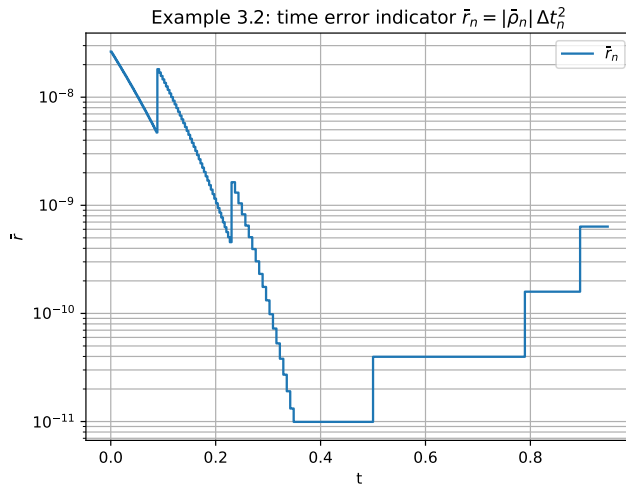


Figure: Error indicators $\bar{r}_n$ for $\varepsilon_{\text{time}} = 1e - 5$, $\varepsilon_{\text{PA}} = 1e - 4$, $\varepsilon_{\delta} = 1e - 10$, $\delta_0 = 0.02$

Step	N	M	$dt_{\min}$	$dt_{\max}$	δ	N.lt	res	η_{time}	η_{PA}	η_{δ} / action
adapt 00	19	3	5.26e-2	5.26e-2	2e-2	6	4.66e-12	5.79e-6/5.26e-7	0/1e-4	1.95e-2/1e-10; refine_time(m=15)
adapt 01	34	3	2.63e-2	5.26e-2	2e-2	3	2.37e-13	1.50e-6/2.94e-7	0/1e-4	1.95e-2/1e-10; refine_time(m=26)
adapt 02	60	3	1.32e-2	5.26e-2	2e-2	3	5.60e-15	3.82e-7/1.67e-7	0/1e-4	1.96e-2/1e-10; refine_time(m=39)
adapt 03	99	3	6.58e-3	5.26e-2	2e-2	2	6.15e-9	9.65e-8/1.01e-7	0/1e-4	1.96e-2/1e-10; $\delta^* = 0.5$
adapt 04	99	3	6.58e-3	5.26e-2	1e-2	4	7.17e-11	1.39e-7/1.01e-7	0/1e-4	9.48e-3/1e-10; refine_time(m=40)
adapt 05	139	3	3.29e-3	5.26e-2	1e-2	2	3.55e-9	3.51e-8/7.19e-8	0/1e-4	9.48e-3/1e-10; $\delta^* = 0.5$
adapt 06	139	3	3.29e-3	5.26e-2	5e-3	4	1.02e-10	4.89e-8/7.19e-8	0/1e-4	4.58e-3/1e-10; $\delta^* = 0.5$
adapt 07	139	3	3.29e-3	5.26e-2	2.5e-3	4	2.29e-10	6.55e-8/7.19e-8	0/1e-4	2.22e-3/1e-10; $\delta^* = 0.5$
adapt 08	139	3	3.29e-3	5.26e-2	1.25e-3	4	2.94e-9	8.27e-8/7.19e-8	0/1e-4	1.08e-3/1e-10; refine_time(m=27)
adapt 09	166	3	1.64e-3	5.26e-2	1.25e-3	3	7.49e-15	2.08e-8/6.02e-8	0/1e-4	1.08e-3/1e-10; $\delta^* = 0.5$
adapt 10	166	3	1.64e-3	5.26e-2	6.25e-4	5	5.27e-14	2.42e-8/6.02e-8	0/1e-4	5.22e-4/1e-10; $\delta^* = 0.5$
adapt 15	166	3	1.64e-3	5.26e-2	1.95e-5	5	1.09e-12	2.64e-8/6.02e-8	0/1e-4	1.44e-5/1e-10; $\delta^* = 0.5$
adapt 16	166	3	1.64e-3	5.26e-2	9.77e-6	5	1.51e-12	2.64e-8/6.02e-8	0/1e-4	7.05e-6/1e-10; $\delta^* = 0.5$
adapt 17	166	3	1.64e-3	5.26e-2	4.88e-6	5	5.07e-12	2.64e-8/6.02e-8	0/1e-4	3.45e-6/1e-10; $\delta^* = 0.5$
adapt 18	166	3	1.64e-3	5.26e-2	2.44e-6	5	1.90e-11	2.64e-8/6.02e-8	0/1e-4	1.69e-6/1e-10; $\delta^* = 0.5$
adapt 19	166	3	1.64e-3	5.26e-2	1.22e-6	5	7.64e-11	2.64e-8/6.02e-8	0/1e-4	8.32e-7/1e-10; $\delta^* = 0.5$
adapt 20	166	3	1.64e-3	5.26e-2	6.10e-7	5	3.61e-10	2.64e-8/6.02e-8	0/1e-4	4.09e-7/1e-10; $\delta^* = 0.5$
adapt 21	166	3	1.64e-3	5.26e-2	3.05e-7	5	2.36e-9	2.64e-8/6.02e-8	0/1e-4	2.01e-7/1e-10; $\delta^* = 0.5$
adapt 22	166	3	1.64e-3	5.26e-2	1.53e-7	6	5.48e-10	2.64e-8/6.02e-8	0/1e-4	9.92e-8/1e-10; $\delta^* = 0.5$
adapt 23	166	3	1.64e-3	5.26e-2	7.63e-8	8	9.00e-9	2.64e-8/6.02e-8	0/1e-4	4.89e-8/1e-10; $\delta^* = 0.5$
adapt 24	166	3	1.64e-3	5.26e-2	3.81e-8	19	4.82e-9	2.64e-8/6.02e-8	0/1e-4	2.42e-8/1e-10; $\delta^* = 0.5$
adapt 25	166	3	1.64e-3	5.26e-2	1.91e-8	40	1.38e-5	2.64e-8/6.02e-8	0/1e-4	1.20e-8/1e-10; $\delta^* = 0.5$
adapt 26	166	3	1.64e-3	5.26e-2	9.54e-9	24	7.73e-9	2.64e-8/6.02e-8	0/1e-4	5.90e-9/1e-10; $\delta^* = 0.5$
adapt 27	166	3	1.64e-3	5.26e-2	4.77e-9	40	7.82e-9	2.64e-8/6.02e-8	0/1e-4	2.92e-9/1e-10; $\delta^* = 0.5$
adapt 28	166	3	1.64e-3	5.26e-2	2.38e-9	40	4.38e-5	2.64e-8/6.02e-8	0/1e-4	1.45e-9/1e-10; $\delta^* = 0.5$
adapt 29	166	3	1.64e-3	5.26e-2	1.19e-9	25	8.07e-9	2.64e-8/6.02e-8	0/1e-4	7.17e-10/1e-10; $\delta^* = 0.5$
adapt 30	166	3	1.64e-3	5.26e-2	5.96e-10	29	9.58e-9	2.64e-8/6.02e-8	0/1e-4	3.55e-10/1e-10; $\delta^* = 0.5$
adapt 31	166	3	1.64e-3	5.26e-2	2.98e-10	40	1.85e-6	2.64e-8/6.02e-8	0/1e-4	1.76e-10/1e-10; $\delta^* = 0.5$
adapt 32	166	3	1.64e-3	5.26e-2	1.49e-10	29	4.07e-9	2.64e-8/6.02e-8	0/1e-4	8.76e-11/1e-10; STOP

$\varepsilon_{\text{time}}$	ε_{PA}	ε_{δ}	$\hat{J}$	J^*	Abs. Error	Relat. Error
1e-5	1e-4	1e-10	4.5316e-5	4.4389e-5	9.269e-7	2.088e-2
1e-4	1e-4	1e-10	5.1801e-5	4.4389e-5	7.412e-6	1.670e-1
1e-3	1e-4	1e-10	7.4538e-5	4.4389e-5	3.015e-5	6.792e-1

$$\hat{J} = \sum_{i=0}^{N-1} \Delta t_i X_i^{10}. \quad (17)$$

$\varepsilon_{\text{time}}$	ε_{PA}	ε_{δ}	E
1e-5	1e-4	1e-10	9.4387e-7
1e-4	1e-4	1e-10	8.2549e-6
1e-3	1e-4	1e-10	4.1346e-5

$$E = \sum_{n=0}^{N-1} \Delta t_n^2 \rho_n \quad (18)$$

Time Refinement

Choose the error tolerance TOL, the initial grid $\{t_n\}_{n=0}^N$, and the parameters s and M , and repeat the following steps:

- 1 Calculate $\{(X_n, \lambda_n)\}_{n=0}^N$ with the symplectic Euler scheme (1.7).
- 2 Calculate error densities $\{\rho_n\}_{n=0}^{N-1}$ and the corresponding approximate error densities

$$\bar{\rho}_n := \text{sgn}(\rho_n) \max(|\rho_n|, K\sqrt{\Delta t_{\max}}).$$

- 3 Break if

$$\max_n \bar{r}_n < \frac{\text{TOL}}{N},$$

where the error indicators are defined by $\bar{r}_n := |\bar{\rho}_n| \Delta t_n^2$.

- 4 Traverse through the mesh and subdivide an interval (t_n, t_{n+1}) into M parts if

$$\bar{r}_n > s \frac{\text{TOL}}{N}.$$

- 5 Update N and $\{t_n\}_{n=0}^N$ to reflect the new mesh.

Constants s and M

Assume that c satisfies

$$\begin{aligned} c &\leq \left| \frac{\bar{\rho}(t)[\text{parent}(n, k)]}{\bar{\rho}(t)[k]} \right| \leq c^{-1}, \quad t \in [t_n^{[k]}, t_{n+1}^{[k]}) \\ c &\leq \left| \frac{\bar{\rho}(t)[k-1]}{\bar{\rho}(t)[k]} \right| \leq c^{-1}, \quad t \in [t_n^{[k]}, t_{n+1}^{[k]}) \end{aligned} \quad (19)$$

for the time steps corresponding to the maximal error indicator on each refinement level and that

$$M^2 > c^{-1}, \quad s \leq \frac{c}{M}. \quad (20)$$

Then, each refinement level either decreases the maximal error indicator with the factor

$$\max_n \bar{r}_n[k+1] \leq \frac{c^{-1}}{M^2} \max_n \bar{r}_n[k]$$

or stops the algorithm.



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